

Module 2

The Profit and Loss Account

Learning Objectives

By the end of this module you should understand:

1. The role of the profit and loss account in the measurement of corporate accomplishment and effort
2. The timing of revenue and expense and the role of accounting conventions
3. The impact on profit of different stock valuation methods and depreciation methods
4. The principal features of depreciation
5. The distinction between product costs and period costs
6. The implications of 'gross profit' and 'net profit'

Agenda

2.1 Introduction

2.2 What Is Profit?

2.3 The Measurement of Accomplishment

2.4 Another Reason for Opting for 'Ship and Invoice'

2.5 Conventions Underlying Measurement of Sales Accomplishment

2.6 The Measurement of Effort

2.7 Task One: Determining the Consumption of the Means of Production

2.8 Task Two: Determining the Value of Closing Work-in-Progress and Inventories

2.9 Types of Inventory in a Manufacturing Company

2.10 Inventory Valuation Methods

2.11 Valuation of Work-in-Progress and Finished Goods

2.12 Interpreting Profit

Review Questions

Over the next two modules we shall be examining the 3 financial statements in greater depth

You Will Become Aware of:

Where the *underlying information* in these statements comes from

The *conventions* employed by accountants in constructing the statements

The various *techniques* used to measure aspects of business activity

We will use the profit and loss account and balance sheet of the sole trader encountered in Module 1 to further explain those three concepts

Profit and Loss Account For the Period Actions 1–7

Sales			€ 750
<i>Less: Cost of sales</i>	Materials	€ 500	
	Labour	20	
	Depreciatio	50	570
Gross Profit			180
<i>Less: Selling and administrative costs</i>			
	Advertising	10	
	Salaries	10	20
Net Profit			€ 160

Fixed assets		
Plant and equipment at cost	€ 12,000	
<i>Less: Depreciation</i>	50	€ 11,950
Current assets		
Inventories	5,500	
Debtors	750	
Cash	4,960	
	11,210	
<i>Less: Current liabilities</i>		
Creditors	3,000	8,210
Net assets of the company		20,160
Represented by:		
Capital introduced	20,000	
Profits earned	€ 160	
Owners' equity		20,160

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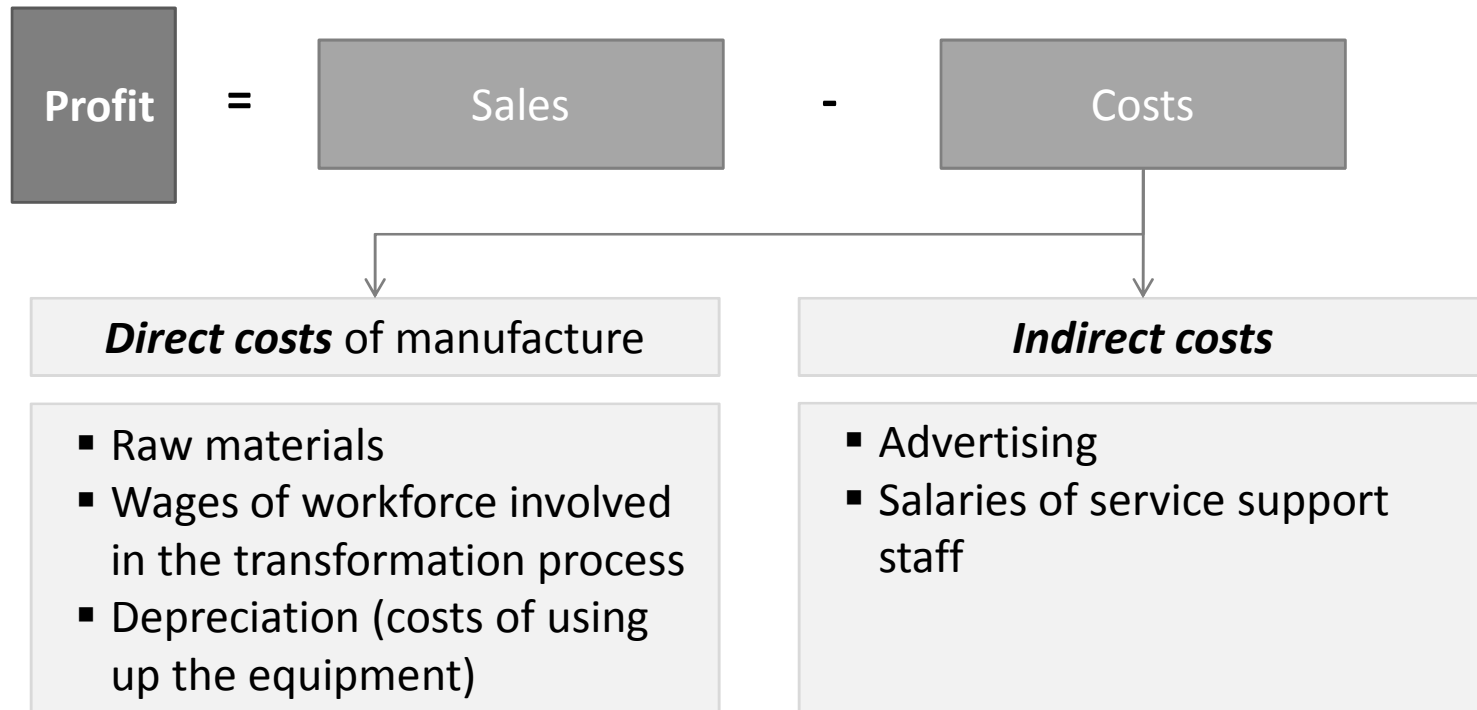
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Review Questions

Profit is the difference between the sales that were made and the sum of costs which have been incurred to bring the goods sold ready for sale



Profit serves as the prime indicator of overall effectiveness

The main purpose of the profit and loss account is to show the results of the transformation process over a given period of time

Transformation Process

The functions that are necessary to convert raw materials into a saleable product.

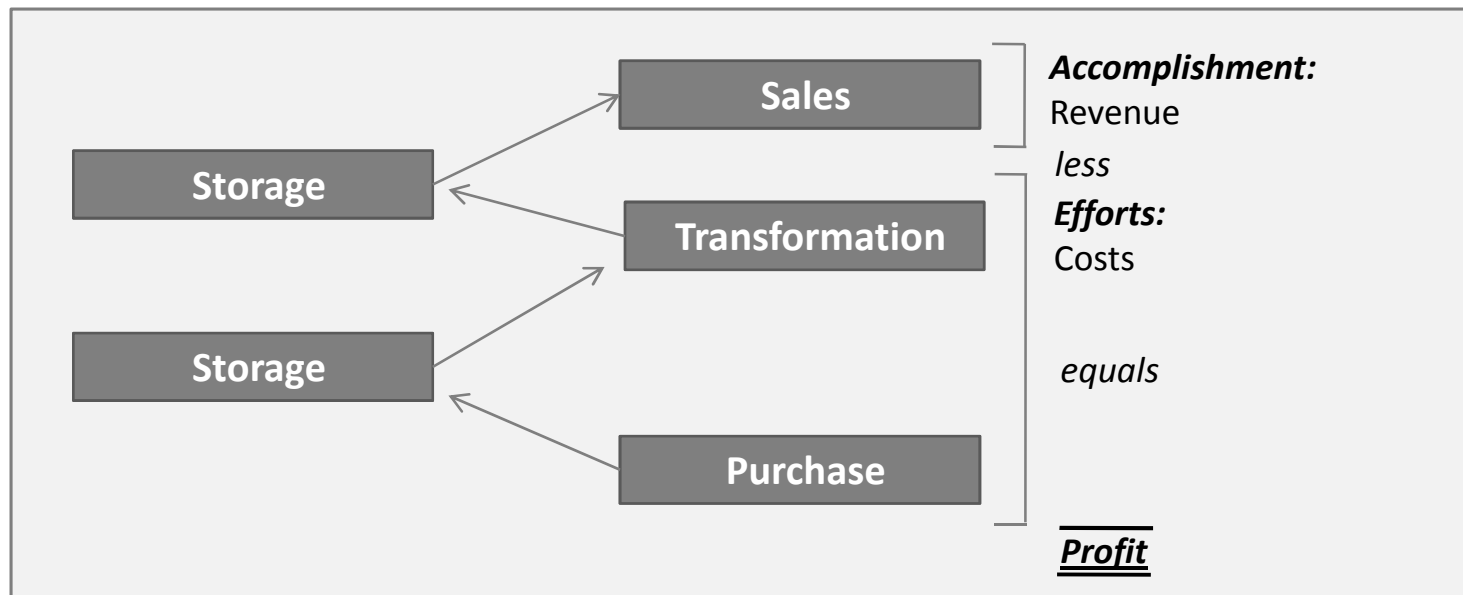


Figure 2.1 Accomplishments less Efforts

Profit

Is the difference between a ***measure of accomplishment*** (what has been achieved in sales) and a ***measure of effort*** (what these sales have cost).

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Review Questions

Products must be SHIPPED and INVOICED to a customer before they are included in the measurement of accomplishment

Example

What can be considered as a *measure of accomplishment* in the transformation process in an accounting period?

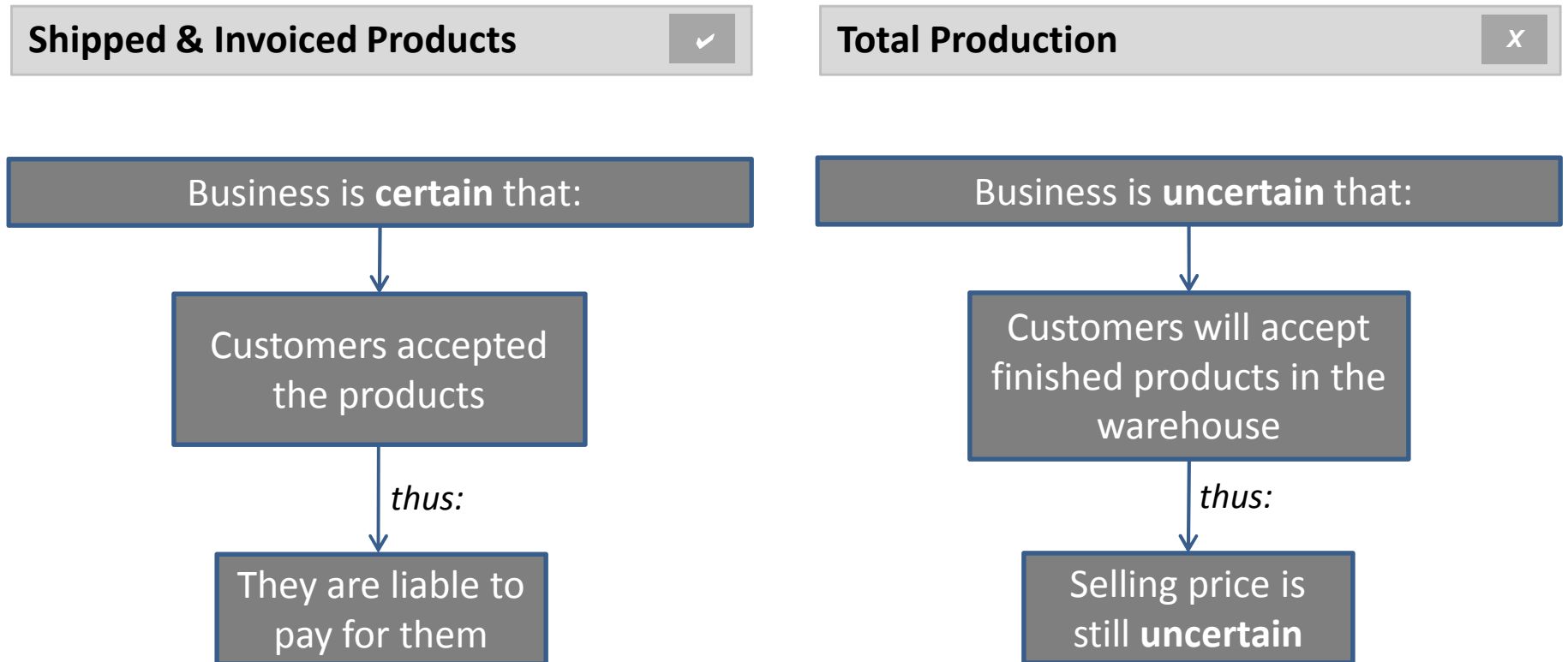
- Total production for one accounting period of Product ABC: 300 units.
- Number of Product ABC shipped and invoiced to customers: 250 units.

The difference between both:

- Products which have been **Produced** but **Not Sold** at the end of the period
- These products are to be stored in the **warehouse**.

The shipped and invoiced products (250 units) are considered as the measurement of accomplishment

The events of shipping and invoicing the products are more critical in measuring accomplishment than total production



Measurement of Accomplishment differs from one company to another according to each one's operation

South African Gold Mine

Occurs at the point when:

The gold is extracted from the rock

They can be **assured** of selling it within a predictable price range

World demand for gold is so buoyant

Ice Cream Seller

Occurs at the point when:

He receives money for ice cream cone

Uncertainties about making an earlier recognition of accomplishment

Accomplishment is generally measured at the first point in the operating cycle at which the 3 following conditions are satisfied

1

The principal revenue-producing service has been performed, i.e. the **product** has been **made and delivered** against a firm **order**.

2

All **costs** that are necessary to create the revenue have either been **incurred** or accurately predictable

3

The **amount collectable** in cash can be **estimated** within an acceptable range of error



Shipping and invoicing is the **most popular** measure of accomplishment

because:

In most cases it is the first point in the cycle at which all **three** criteria are met

Three other methods for measuring accomplishment exist – the first is time of sale orders in which sales are recorded as soon as orders are received...

Time of Sales Orders

- Useful in situations where:

1

The goods which are ordered are in **inventory** when the order is received

2

All major **costs** have already been incurred

3

The number and timing of cancelled orders do **not** vary significantly

Example

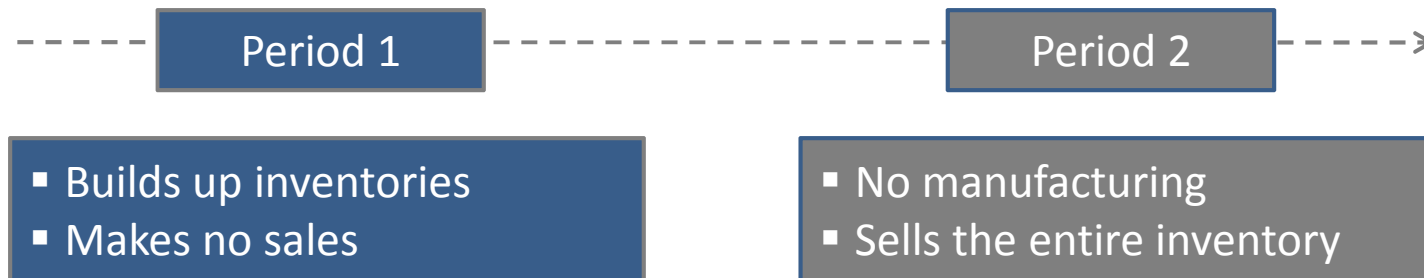
A knitwear company received an order to supply high-quality sweaters to the Britain/ Europe Ryder Cup golf team for their next match against the USA.

- A firm order? **Yes.**
- A customer who will pay? **Yes.**
- Certain production costs? **No.** The world price of cashmere is volatile and may change significantly between order and delivery.

...The second measure of accomplishment is time of production where sales are recorded as soon as products have been produced...

Time of Production

Consider this manufacturing company:



Recognizing accomplishment in period 1

x

- Inventory of finished goods in the balance sheet for period 1
- Cost of sales in the profit and loss account for period 2

✓

Note

In some industries like shipbuilding, civil engineering and construction, the time of production is acceptable when sales are certain following production

Using the timing of production method of measuring accomplishment allows some companies to record profits when they occur

Example

XY Construction plc is in the process of building a bridge linking two Scottish Hebridean islands. The job will take two years.

- At the end of the first year, one-third of the contract of €12 million had been completed, although the client had paid only one-sixth, namely, €2 million.
- The company expected to make €3 million profit on the bridge and at the end of Year 1 costs were being incurred on a predictably even basis.
- How many sales and profits should XY recognize at the end of Year 1?

Solution

Year 1

P&L Account

Sales €4 Mio	-	Costs €3 Mio	=	Profit €1 Mio
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Balance Sheet

Debtors + €2 Mio

Year 2

P&L Account

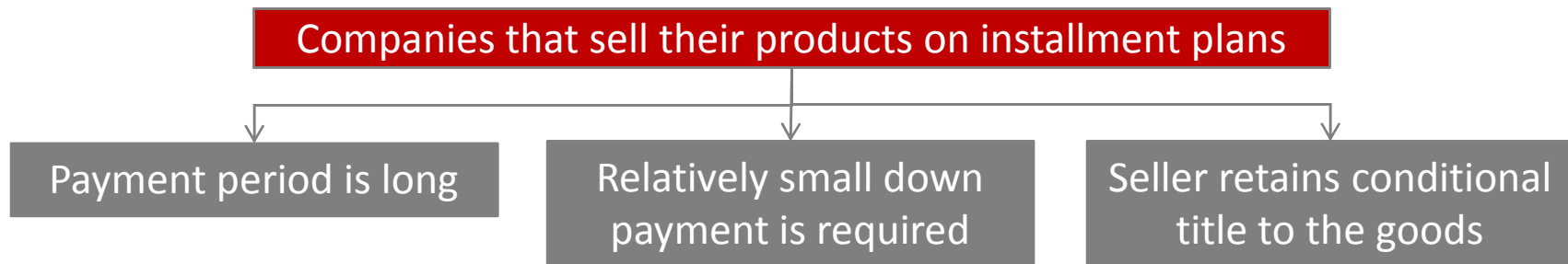
Sales €8 Mio	-	Costs €6 Mio	=	Profit €2 Mio
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A third method is time of collection in which accomplishment is recognized only when payment has been collected

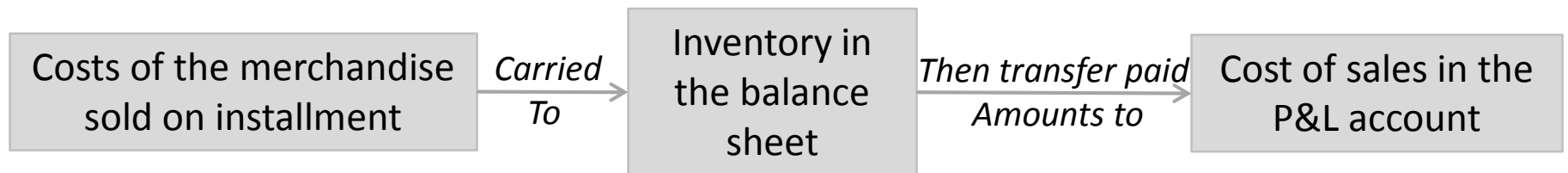
Why Is Time of Collection Sometimes a Critical Point in Measuring Accomplishments?

It is a critical when:

- The likelihood of receiving payment **cannot** be predicted accurately
- There is a relatively **high risk** of reclaims of the merchandise because of non-payment.



Here, revenue is recognized as the cash is collected



Accomplishments are conveniently measured when the products are invoiced unless there is a likelihood that the customers would not pay

When Is a Sale a Sale?

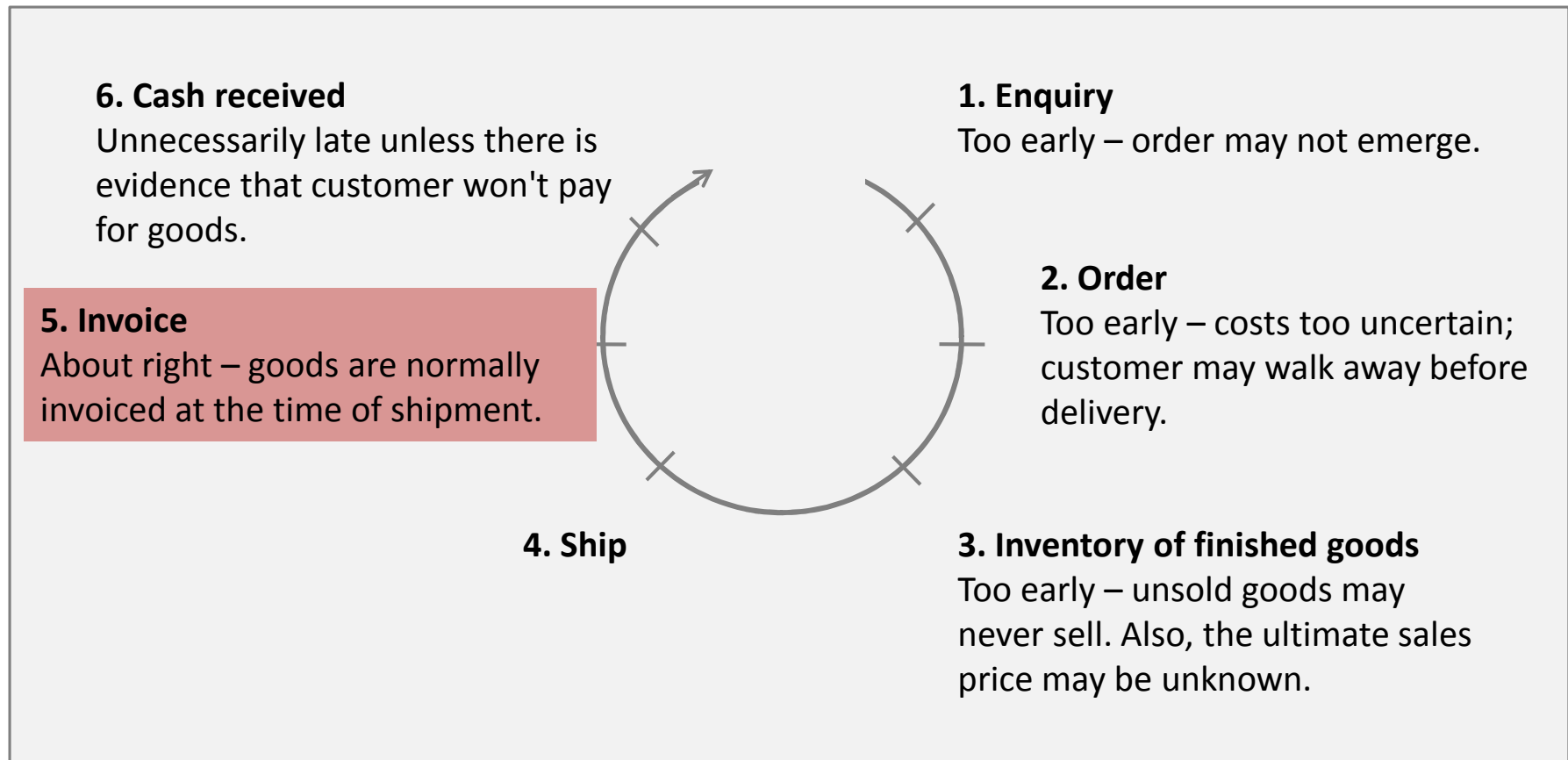


Figure 2.2 A typical manufacturing cycle

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Review Questions

Let's go back and take a look at the example from module 1

Sales			€ 750
Less: Cost of sales	Materials	€ 500	
	Labour	20	
	Depreciatio	50	570
Gross Profit			180
Less: Selling and administrative costs			
	Advertising	10	
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Fixed assets		
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Current assets		
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Cash	4,960	
	11,210	
Less: Current liabilities		
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Net assets of the company		20,160
Represented by:		
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All sales were on credit, he received **NO CASH** for them at the time of sale and until the end of the period.

It makes sense in this example to record sales as soon as they are invoiced since the business is fairly certain of receiving payment for them

- On the understanding that the trader is confident that all customers would pay for the goods within the agreed credit period (e.g. 30 days after sale), thus:

The business recognizes the full figure of €750 as the measurement of accomplishment at the end of Action 7, that is the outcome of the economic effort which has been expended during the period.

- Were the business to delay recognition of sales to the next period when the cash would have been received, then the measurement of accomplishment in the second period would be a mixture of:
 - a) Sales made in the second period and paid for in the second period.
 - b) Cash of €750 received for sales made in the first period.

Thereby failing to measure accomplishment correctly in either of the two periods.

We will discuss another example to further demonstrate why we should choose shipped and invoiced products as the measure of accomplishment

Problem

A manufacturer of sports bags supplies major retail sports outlets. The relevant figures for production and sales for one year are as follows:

Orders received during the year by the manufacturer's sales force	15,000
Sports bags manufactured	10,000
Sports bags shipped and invoiced	12,000
Faulty bags returned and not yet repaired	1,000

How many sports bags were sold during the year?

15,000	10,000	12,000
Cannot be used	Cannot be used	Can be used
Orders are not satisfied	2,000 bags have been drawn from inventory to satisfy the order	Must be reduced by 1,000 faulty bags that were returned to inventory
Customers can walk away		

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Review Questions

The accounting series of techniques and procedures developed by accountants are called conventions; there are 2 accounting conventions for measuring sales

Two Conventions are Adopted in Measuring The Sales Figure:

The realization Convention

**Product delivered /
Service performed**

Sales measured as:

Only products that have been sold

The accruals Convention

Cash does **not have to be received**

Sales measured as:

Any product which is completed or partially completed

An order from a creditworthy customer

An obligation to a supplier

The accruals convention covers situations where entities pay for products covering periods stretching beyond the dates of their financial statements

Examples

1. If a sole trader pays his annual insurance premium on his assets on 1 October 20x1 for the following 12 months

At the end of the year the P&L account would include only 3 months' worth of this premium (and presumably 9 months' worth of the premium paid on 1 October 20x0), and the balance sheet would record the 9 months' balance of premium not recognized in the profit and loss account as a current asset called 'Prepayments'.

Examples

2. When the sole trader pays his property tax in arrears annually on 1 March 20x2, he would accrue ten months' worth of the expected payment at 31 December 20x1.

That is, he would pretend he had paid it so that his profit and loss account for the year to 31 December 20x1 would reflect the time consumption of resource in the 12 months of the accounts and the balance sheet would show this ten months' unpaid tax as a current liability called 'Accruals'.

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Review Questions

Efforts are all the costs included in producing saleable products

What Is Meant By 'Efforts'?

Efforts are all the costs involved in producing saleable products, and the costs involved in actually selling the products which are recognized in the measurement of sales accomplishment.

Some typical costs of production:

- Raw materials
- Labor
- Depreciation of production plant and machinery
- Overheads directly related to the production cycle

Some typical costs of selling and administration:

- Advertising and promotion
- Packaging and distribution
- Staff salaries
- Non-production overhead expenditure
- Research and development

The measurement of effort is governed by three conventions, namely the matching convention, the allocation convention and the cost convention

Matching Convention

Profit is arrived at:

By matching the effort (**Costs**) with the units shipped and invoiced to the customers (**Sales**) during the period.

Allocation Convention

First task

Is to determine how much of each means of production was **consumed** during the accounting period

Second task

Is to determine how much of each means of production should be matched with sales revenue and how much should be added to the work-in-progress and to the inventory of finished goods (matching)

Cost Convention

Using the historical **acquisition cost** of different means of production

The matching convention is very useful in situations where sales volume does not equal production volume

Sales Volume > Production Volume

As in the sports bag example but ignoring faulty goods

	Bags
Sales	12,000
Cost of Sales	
a) Cost of Manufacturing	10,000
Plus:	
b) Costs attaching to bags taken from finished goods inventory	2,000
	12,000

Production Volume > Sales Volume

Reverse the numbers of the sports bag example, i.e. sold 10,000, made 12,000

Sales	10,000
Cost of Sales	
a) Cost of Manufacturing	12,000
Less:	
b) Costs attaching to bags taken from finished goods inventory	2,000
	10,000

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Review Questions

The first task of the allocation convention involves determining the consumption of means of production in terms of raw material, labor and depreciation

Raw Material

"Often the largest of all the costs of production"



Physical quantity of raw materials at the **end** of the year

-

Sum of **opening** inventory of raw materials and purchases during the year

Labor

The annual (or monthly, or weekly) **payroll** is the usual source for the measurement of labor effort expended

Depreciation

The decline in value of fixed assets is an expense

Depreciation charges are based on **procedures** devised by engineers and accountants trying to reflect the periodic use of plant and machinery

Straight-line method

Reducing Balance
Depreciation

Consumption method

An example showing how to determine raw material at the end of the accounting period

Example

XY Ltd commenced the year with 100 units of raw materials. During the year the company purchased 1,200 units. At the end of the year 300 units were left in the raw material inventory. The consumption of raw materials was therefore 1,000 units. Physical units can normally be counted relatively easily and the inventories can be calculated at the beginning and at the end of the year from the purchase invoices. Greater problems arise when we start to value the amount consumed. Assume that the 1,200 units were purchased in a period of rising raw material prices at two-monthly intervals as follows:

February	200 Units	@ €1 each	€200
April	200 Units	@ €1.5 each	300
June	200 Units	@ €2 each	400
August	200 Units	@ €2.5 each	500
October	200 Units	@ €3 each	600
December	200 Units	@ €3.5 each	700
	1,200 Units		2,700

An example showing how to determine raw material at the end of the accounting period (Continued)

Raw Material “Opening Inventory”

The opening inventory of 100 units was valued at €1 per unit. Therefore the total costs to be allocated amount to €2,800

Raw Materials “Closing Inventory”

4 valuation possibilities suggest themselves:

Valuation Method	Number of Units	Price	Closing Stock Valuation	Valuation of units consumed	Total	
1	300	@3.5	€1,050	€1,750	€2,800	Last price paid for all the closing inventory
2	200	@3.5	1,000	1,800	2,800	Counting back using actual price
	100	@3				
3	300	@1	300	2,500	800	Using earliest prices
4	300	@2.15	645	2,155	2,800	Using the average cost of units $€2,800 \div 1,300$

The difference in stock valuation is significant, therefore there is a difference in cost of units consumed

Depreciation is critical in determining the consumption of means of production as the decline in value of fixed assets is considered an expense

Depreciation

The allocation of the cost of assets purchased in one accounting period over the accounting periods in which they are used

Three Important Factors To Be Considered When Calculating Depreciation:

Acquisition Cost

The actual historic cost, including installation charges if any.

Estimated net residual amount

The value the business will receive for the asset on disposal.

Useful life

The estimated useful life of the asset to the present owner.

Two out of the three factors are nothing more than estimates

Although there are a lot of methods to calculate depreciation, only the most common three will be discussed here

Three Methods of Depreciation will be Illustrated Using the Following Example:

Example

Machine X

Acquisition Costs		€1,250	
Estimated net residual value		50	
Estimated service life		3	Years
Estimated pattern of running hours:	Year 1	5,000	Hours
	Year 2	8,000	Hours
	Year 3	2,000	Hours

The straight-line depreciation method is widely used because it is simple and reasonably applicable to many kinds of fixed assets

1. The Straight-line Method

$$\begin{aligned} \text{Annual depreciation charge} &= \frac{\text{Acquisition Cost} - \text{Estimated Residual Value}}{\text{Estimated Service Life}} \\ &= \frac{€1,250 - €50}{3} = €400 \end{aligned}$$

Depreciation Schedule:

Year	Depreciation Expense	End of Year	
		Balance in Accumulated Depreciation	Book Value
At Acquisition			€1,250
1	€400	€400	850
2	400	800	450
3	400	1,200	50
	€1,200		

This method is more appropriate when the use of the asset is the same each year & its usefulness is declining evenly over its life

Reducing balance depreciation reports large depreciation expense in earlier years followed by reduced depreciation expenses

2. Reducing Balance Depreciation

$$\text{Annual Depreciation Charge} = \left(1 - \sqrt[n]{\frac{\text{Scrap Value}}{\text{Cost}}} \right) \times \text{Cost}$$

n = Estimated service life of the asset

$$= \left(1 - \sqrt[3]{\frac{50}{1,250}} \right) \times 1,250 = 66\%$$

Depreciation Schedule:

Year	Depreciation Expense	End of Year	
		Balance in Accumulated Depreciation	Book Value
At Acquisition			€1,250
1	€825	€825	425
2	280	1,105	145
3	95	1,200	50
	€1,200		

**Rationale: asset is more efficient in generating revenue in the earlier years
AND repair expenditure tends to be low in the early years**

Consumption method is based on the number of running hours of the machine

3. Consumption Method

$$\text{Annual depreciation charge} = \frac{\text{Annual running hours}}{\text{Total number of running hours}} \times \text{Net Cost}$$

$$\text{Year 1} \quad 5,000/15,000 \times \text{€}1,200 = \text{€}400$$

$$\text{Year 2} \quad 8,000/15,000 \times \text{€}1,200 = \text{€}640$$

$$\text{Year 3} \quad 2,000/15,000 \times \text{€}1,200 = \text{€}160$$

Depreciation Schedule:

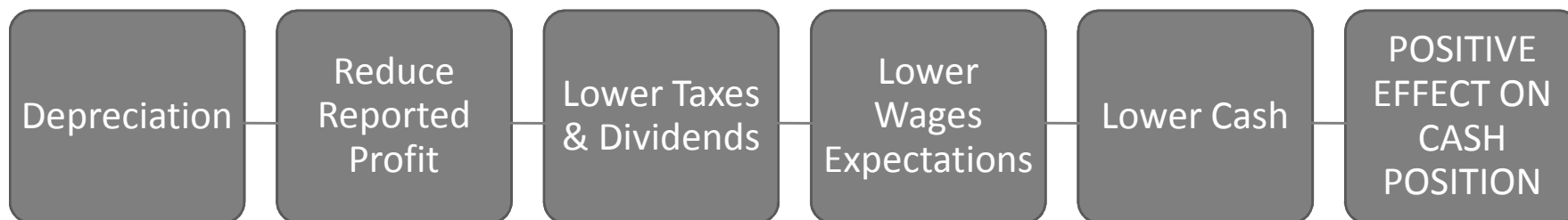
Year	Depreciation Expense	End of Year	
		Balance in Accumulated Depreciation	Book Value
At Acquisition			€1,250
1	€400	€400	850
2	640	1,040	210
3	160	1,200	50
	€1,200		

The selection of depreciation method influences the amount of cost allocated to each accounting period which influences reported profit

Comparison of Methods

Year	Straight-line Method	Depreciation Expense	
		Reducing balance	Consumption method
1	€400	€833	€400
2	400	278	640
3	400	93	160
	€1,200	(say) €1,200	€1,200

Relationship Between Depreciation, Profit and Cash Position



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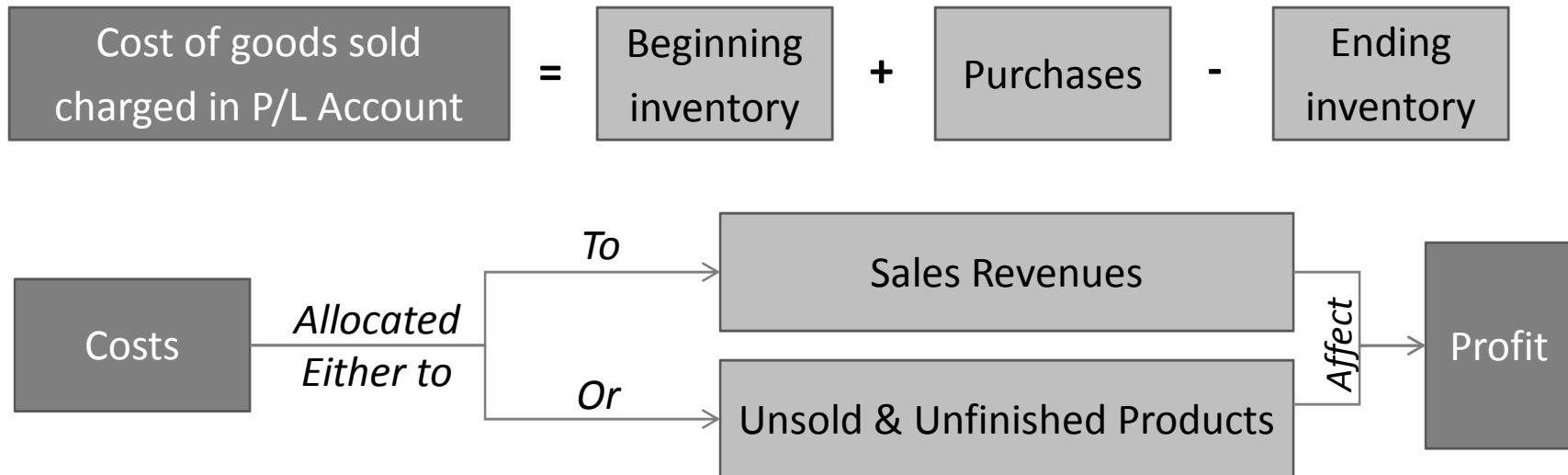
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Review Questions

To measure the efforts involved in the production, the various means of production consumed during the accounting period must first be measured



- The **higher** the ending inventory valuation, the **higher** the reported profit
- The **lower** the ending inventory valuation, the **lower** the reported profit

The selection of inventory valuation system is very important since it impacts the measurement of effort and consequently the profit

The higher the ending inventory the higher the profit

Example

Refer back to the example in Task One on raw materials. Here we saw four inventory valuations being put forward, each one different. Let us imagine the 1,000 units sold realized €5 each; Sales (measurement of accomplishment) would therefore amount to €5,000. Consider the four profit and loss accounts that would emerge using four inventory valuation methods:

	1	2	3	4
Sales	€5,000	€5,000	€5,000	€5,000
Cost of Sales	1,750	1,800	2,500	2,155
Profit	€3,250	€3,200	€2,500	€2,845
Closing Inventory	€1,050	€1,000	€300	€645

Method 1 reports the highest profit among the other valuation methods

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Review Questions

In a manufacturing operation we usually find 3 inventory categories

Inventory Categories



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Review Questions

Inventory is valued at the lower of: cost OR net realizable value

	Cost Convention	Conservative Convention
When it is used	Generally Used	When Selling Price < Cost
Inventory valuation	Inventory is valued in terms of cost	Inventory is valued at net realizable value
Comment	Sum of direct and indirect expenditures incurred in producing saleable goods	Used when it is wrong to value inventory at cost ▪ Understating cost of sales ▪ Overstating income

Accountants therefore adopt the more conservative rule

Inventory is valued at the lower of cost or net realizable value

An example of the conservatism convention over-riding the cost convention

Example

Refer back again to the example in Task One and select valuation method 2; we see that 1,000 units sold for €5,000, cost of sales were €1,800, profit was €3,200 and closing inventory was €1,000, that is, 300 units of which 200 were valued at €3.50 and 100 at €3.00. This valuation of €1,000 is based on the cost convention.

Let us assume that a technological advance announced towards the end of the period rendered the closing inventory obsolete and the business considered that only €100 would be raised from its sale as scrap.

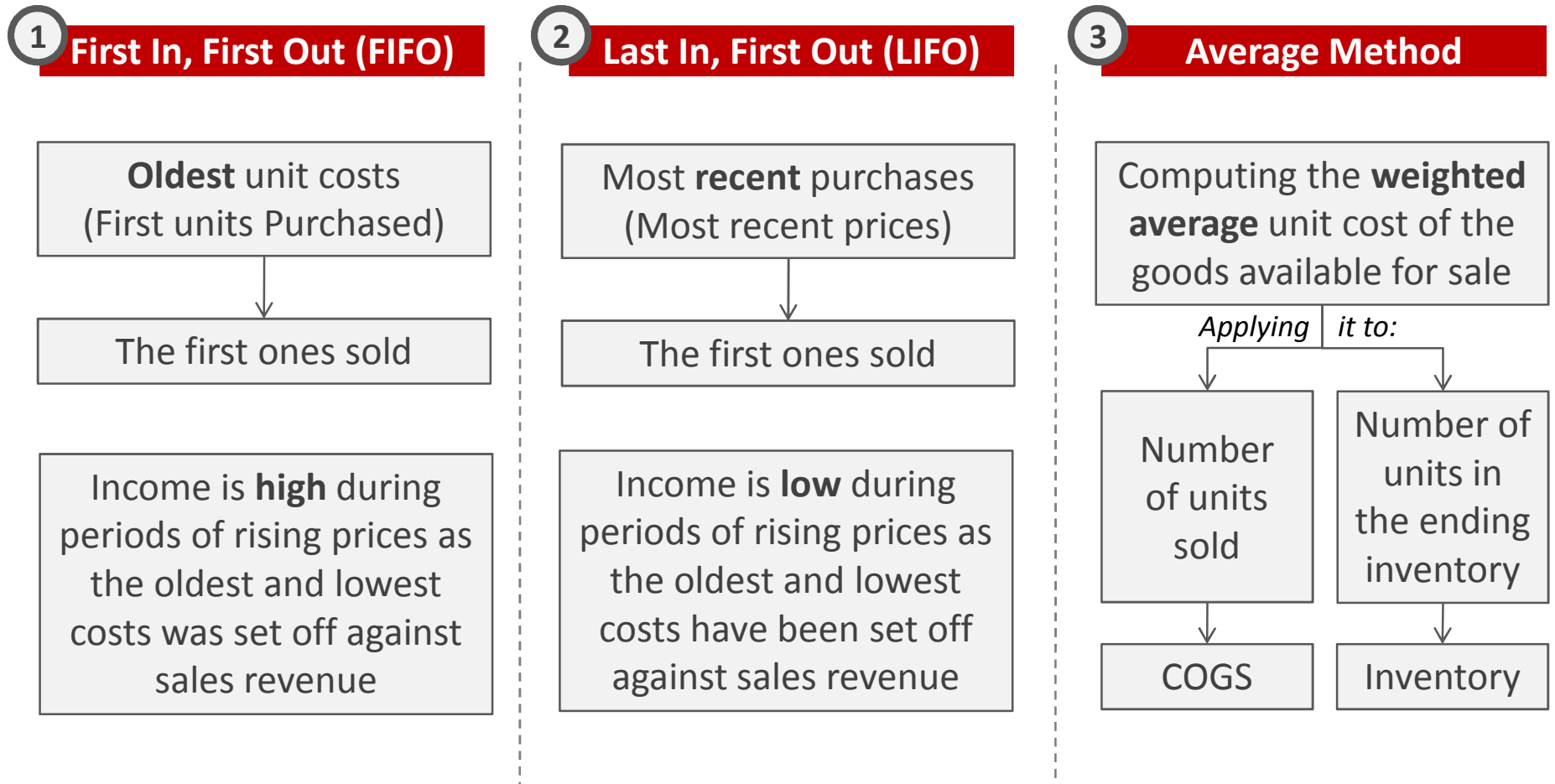
Now the profit and loss would read:

	€
Sales	€5,000
Cost of Sales (1,800 + 900)	2,700
Profit	€2,300
Closing Inventory (1,000 – 900)	100

If the adjustments were not to be made in this period, the €900 additional costs would have to be written off in the next period

Conservatism says, 'Recognize losses and reductions in value whenever they come to light'

There are many inventory costing methods; the three most important ones are FIFO, LIFO and the Average Method



An example showing how to determine raw material at the end of the accounting period

Example

XY Ltd commenced the year with 100 units of raw materials. During the year the company purchased 1,200 units. At the end of the year 300 units were left in the raw material inventory. The consumption of raw materials was therefore 1,000 units. Physical units can normally be counted relatively easily and the inventories can be calculated at the beginning and at the end of the year from the purchase invoices. Greater problems arise when we start to value the amount consumed. Assume that the 1,200 units were purchased in a period of rising raw material prices at two-monthly intervals as follows:

February	200 Units	@ €1 each	€200
April	200 Units	@ €1.5 each	300
June	200 Units	@ €2 each	400
August	200 Units	@ €2.5 each	500
October	200 Units	@ €3 each	600
December	200 Units	@ €3.5 each	700
	1,200 Units		2,700

An example showing how to determine raw material at the end of the accounting period (Continued)

Raw Materials “Closing Inventory”

4 valuation methods suggest themselves:

Valuation Method	Number of Units	Price	Closing Stock Valuation	Valuation of units consumed	Total	
1	300	@3.5	€1,050	€1,750	€2,800	Not Recognized
2	200	@3.5	1,000	1,800	2,800	
	100	@3				FIFO
3	300	@1	300	2,500	800	
4	300	@2.15	645	2,155	2,800	LIFO
						Average Method

Most managers believe that the best inventory valuation method is the one that best matches the sales pricing policy of the company

No one system can be considered ‘best’. Management should choose the system which suits their long-term goals

Agenda

2.1 Introduction

2.2 What Is Profit?

2.3 The Measurement of Accomplishment

2.4 Another Reason for Opting for 'Ship and Invoice'

2.5 Conventions Underlying Measurement of Sales Accomplishment

2.6 The Measurement of Effort

2.7 Task One: Determining the Consumption of the Means of Production

2.8 Task Two: Determining the Value of Closing Work-in-Progress and Inventories

2.9 Types of Inventory in a Manufacturing Company

2.10 Inventory Valuation Methods

2.11 Valuation of Work-in-Progress and Finished Goods

2.12 Interpreting Profit

Review Questions

The other two categories of inventories – work-in-progress and finished goods- must now be examined

We have to consider:

Product Costs

Costs involved in converting raw materials into saleable goods

Period Costs

Costs incurred and set off against sales revenue in the P/L Account and not included in the valuation of closing inventory

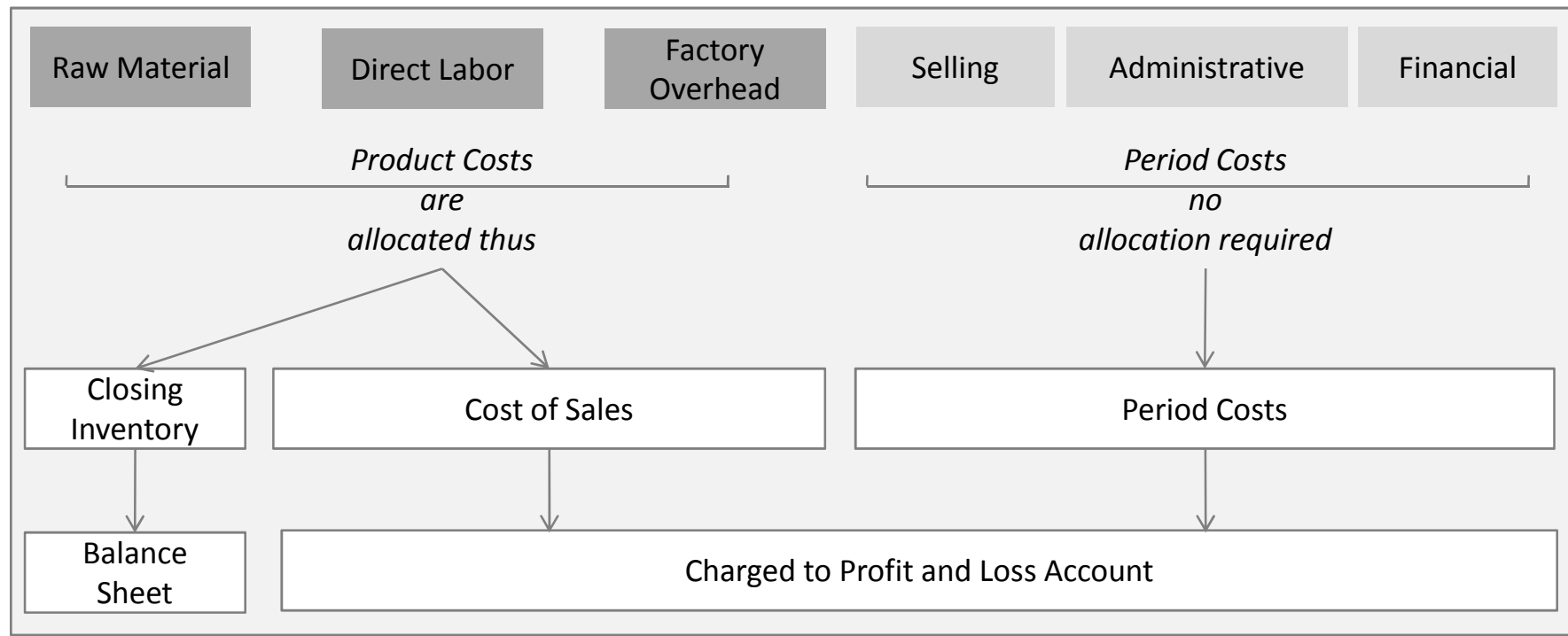
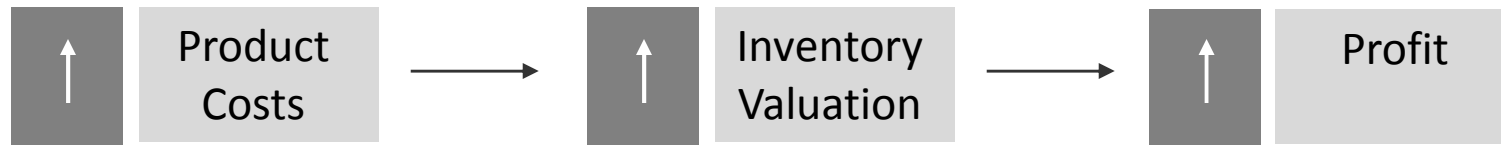


Figure 2.4 Allocation of product and period costs

The way in which a company classifies its costs into period costs and product costs can have a significant effect on the reported profit figure



Some accountants argue that a portion of administrative costs should be considered as a product cost.

Accountants do not agree where the line should be drawn between product costs and period costs

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Review Questions

Great care must be taken in interpreting the profit or loss figure

Extended Profit and Loss Account for Sole Trader for the Period Ended Action 7

Sales		750
Less: Cost of sales:		
Materials	500	
Labour	20	
Depreciation	50	570
Gross profit		<u>180</u>
Advertising	10	
Salaries	10	20
Net profit before interest and taxes		<u>160</u>
Interest charges		10
Net profit after interest but before taxes		<u>150</u>
Tax		50
Net profit after interest and taxes		<u><u>€100</u></u>

Since there are different ways to measure accomplishment and effort, we have to measure profit in a consistent manner from one period to another

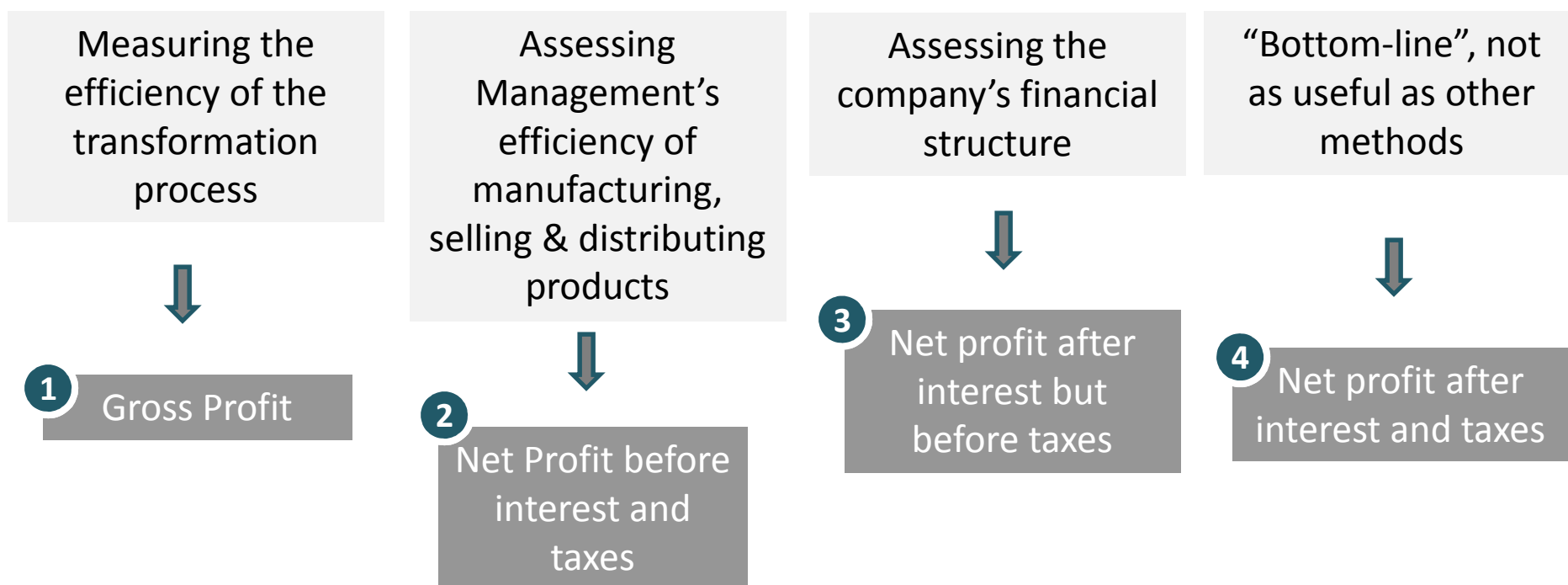
It should be clear for what purpose the profit figure is intended...

Extended Profit and Loss Account for Sole Trader for the Period Ended Action 7

	Sales		750
	Less: Cost of sales:		
	Materials	500	
	Labour	20	
	Depreciation	50	570
1	Gross profit		180
	Advertising	10	
	Salaries	10	20
3	Net profit before interest and taxes		160
	Interest charges		10
4	Net profit after interest but before taxes		150
	Tax		50
2	Net profit after interest and taxes		€100

...And accordingly we would select the profit measure we need to interpret

If we seek:



Net profit (4) after interest and taxes tells very little about the company's efficiency and does not yield sufficient information for a judgment to be made on managerial efficiency.

Agenda

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Review Questions

Multiple Choice Questions

2.1 In the profit and loss account, which of the following is regarded as the measure of accomplishment?

- A. Total manufacturing output.
- B. Total sales orders received.
- C. Total shipped and invoiced sales.
- D. Total cash receipts from debtors.

Multiple Choice Questions (Continued)

2.2 In the profit and loss account, which of the following accounting conventions is adopted in measuring the sales figure in the profit and loss account?

A. Conservatism.

B. Accruals.

C. Matching.

D. Cost.

Multiple Choice Questions (Continued)

2.3 In the profit and loss account, which of the following accounting conventions is applied in the measurement of effort in the profit and loss account?

- A. Matching.
- B. Realization.
- C. Accruals.
- D. Going concern.

Multiple Choice Questions (Continued)

2.4 Which of the following is a typical cost of selling and administration?

- A. Depreciation of plant and machinery.
- B. Wages of production staff.
- C. Repairs to plant and machinery.
- D. Packing and distribution.

Multiple Choice Questions (Continued)

2.5 Which of the following is a typical cost of production?

- A. Salaries of office staff.
- B. Product research and development.
- C. Raw materials.
- D. Expenditure on brochures.

Multiple Choice Questions (Continued)

The following information applies to Questions 2.6–2.12.

At 31 December, James White & Co carried out its annual physical stocktaking, counting 25 color television sets, each of which had cost €100. During the next year, further purchases were made as follows.

Month	Sets purchased	Unit price
March	5	€100
May	10	€110
June	8	€115
September	6	€120
November	12	€125

Annual sales totalled 52 sets, amounting to €9350.

2.6 What is the amount of purchases during the year?

A. €2,500.

B. €4,740.

C. €6,850.

D. €7,240.

Multiple Choice Questions (Continued)

2.7 What is the total cost to be allocated between closing inventory and cost of sales?

A. €2,240.

B. €4,759.

C. €7,240.

D. €9,940.

Opening inventory ($25 \times €100$)	2 500
<i>Add: Purchases</i>	4 740
	7 240

Multiple Choice Questions (Continued)

2.8 If the first-in, first-out (FIFO) method of inventory valuation is used, what is the amount of closing inventory?

A. €1,740.

B. €1,750.

C. €2,500.

D. €5,500.

Opening inventory	25 sets
Add: Purchases	41
	66
Less: Sales	52
Closing inventory	14

Using FIFO, the approach is to count back, using actual prices, from the last purchase.

November – 12 sets at €125	€1 500
September – 2 sets at €120	€ 240
November – 14	€1 740

Multiple Choice Questions (Continued)

2.9 If the first-in, first-out (FIFO) method of inventory valuation is used, what is the amount of cost of sales?

- A. €1,740.
- B. €2,500.
- C. €5,500.
- D. €7,240.

Opening inventory	€2 500
<i>Add:</i> Purchases	4 740
	7 240
 <i>Less:</i> Closing inventory	 1 740
Cost of sales	
	€5 500

Multiple Choice Questions (Continued)

2.10 If the average cost method of inventory valuation is used, what is the cost per set in closing inventory?

A. €109.70

B. €114.00

C. €120.60

D. €122.50

Opening inventory	25 at €100	€2 500
Add: Purchases	41	4 740
	<u>66</u>	<u>€7 240</u>

$$\text{Average cost} = \frac{€7240}{66} = €109.70$$

Multiple Choice Questions (Continued)

2.11 If the last-in, first-out (LIFO) method of inventory valuation is used, what is the amount of closing inventory?

A. €1,400.

B. €1,470.

C. €1,540.

D. €1,680.

Multiple Choice Questions (Continued)

2.12 Which of the following methods of inventory valuation will result in the highest cost of sales?

- A. Weighted-average cost.
- B. Last-in, first-out (LIFO).
- C. Average cost for period purchases.
- D. First-in, first-out (FIFO).

Multiple Choice Questions (Continued)

2.13 Which of the following factors in depreciation computations is based on an actual figure?

- A. Residual value.
- B. Useful life of asset.
- C. Replacement value.
- D. Acquisition cost.

Multiple Choice Questions (Continued)

The following information applies to Questions 2.14–2.17.

JK Builders Co purchases a new excavator, costing €40,000. Its expected useful lifetime is 10 years, at which point it is anticipated that it will be capable of resale for €6,000.

2.14 If the straight-line method of depreciation is used, what is the annual depreciation charge?

A. €2,800.

B. €3,400.

C. €4,000.

D. €4,600.

$$\begin{aligned}\text{Annual charge} &= \frac{\text{Cost} - \text{Residual value}}{\text{Useful life}} \\ &= \frac{€40\,000 - €6\,000}{10} = €3\,400\end{aligned}$$

Multiple Choice Questions (Continued)

2.15 If the straight-line method of depreciation is used, what is the book value of the excavator at the end of four years?

- A. €20,400.
- B. €22,800.
- C. €26,400.**
- D. €29,900.

Year	Depreciation charge	Accumulated depreciation	Book value
1	3 400	3 400	36 600
2	3 400	6 800	33 200
3	3 400	10 200	29 800
4	3 400	13 600	26 400

Multiple Choice Questions (Continued)

2.16 If the reducing balance method of depreciation is used, requiring an annual depreciation rate of 15 per cent, what is the annual depreciation charge in Year 2?

- A. €3,400.
- B. €4,335.
- C. €5,100.
- D. €6,000.

Year	Depreciation charge	Accumulated depreciation	Book value
1	6 000	6 000	34 000
2	5 100	11 100	28 900
3	4 335	15 435	24 565
4	3 685	19 120	20 880

Multiple Choice Questions (Continued)

2.17 If the reducing balance method of depreciation is used, what is the book value of the excavator at the end of three years?

A. €16,000.

B. €18,905.

C. €22,400.

D. €24,565.

Multiple Choice Questions (Continued)

2.18 In considering its features, it is important to appreciate that depreciation:

- i. produces cash for fixed asset replacements;
- ii. is an allocation of the cost of fixed assets over their period of use;
- iii. is a part of the cost of financial charges;
- iv. is a measure of the value of the consumption of a fixed asset.

Which of the following is correct?

- A. (i) and (ii) only.
- B. (i) and (iii) only.
- C. (ii) and (iii) only.
- D. (ii) and (iv) only.

Multiple Choice Questions (Continued)

2.19 In valuing closing inventory at net realizable value, which of the following accounting conventions is of prime importance?

A. Conservatism.

B. Realization.

C. Matching.

D. Cost.

Multiple Choice Questions (Continued)

2.20 Which of the following is a period cost?

- A. Maintenance engineers' wages.
- B. Financial charges.
- C. Heat, light and power.
- D. Depreciation of plant.

Multiple Choice Questions (Continued)

2.21 Which of the following is a factory overhead?

- A. Accountants' salaries.
- B. Insurance on sales vehicles.
- C. Foremen's wages.
- D. Bank charges.

Multiple Choice Questions (Continued)

2.22 Which of the following profit measures is appropriate to use in assessing management's efficiency in manufacturing, selling and distributing its products?

- A. Net profit before interest and taxes.
- B. Net profit after interest and taxes.
- C. Net profit after interest, but before taxes.
- D. Gross profit.

Multiple Choice Questions (Continued)

2.23 What is the total amount of extra depreciation at the end of 20x4 to introduce a replacement cost-based depreciation charge?

- A. €1,100.
- B. €1,320.
- C. €2,200.
- D. €3,300.

$$\begin{aligned}\text{Replacement cost of fixed assets in 20x4} &= \text{€110 000} \times 1.10 \\ &= \text{€121 000}\end{aligned}$$

Historic cost – annual depreciation	€11 000
Replacement cost – annual depreciation	12 100
Extra depreciation	<u>1 100</u>

Multiple Choice Questions (Continued)

2.24 What is the amount of 'top-up' depreciation at the end of 20x5 for a replacement cost-based depreciation charge?

- A. €1,100.
- B. €1,200.
- C. €1,210.
- D. €2,310.

$$\begin{aligned}\text{Replacement cost of fixed assets in 20x5} &= €110\,000 \times 1.10 \times 1.10 \\ &= €133\,100\end{aligned}$$

Therefore, revised annual charge on replacement cost basis is €13 310 for 20x5. To 'top up' the 20x4 depreciation charge (on replacement cost basis) of €12 100 – as in 23 above – an extra amount of €1210 will be required.

Multiple Choice Questions (Continued)

The following information applies to Questions 2.25–2.27.

Hare and Howndes Limited has prepared its profit and loss account, on a historic cost basis, for the year ending 31 December 20x5.

	€	€
Sales		6 250
Less: Cost of sales		
Opening inventory	600	
Purchases	3 250	
	<u>3 850</u>	
Closing inventory	790	3 060
Gross profit		<u>3 190</u>

The trade index relating to materials supplied to Hare and Howndes is:

1 January 20x5: 100

31 December 20x5: 130

Multiple Choice Questions (Continued)

2.25 Using the averaging method, what is the revised opening inventory?

A. €522.

B. €600.

C. €690.

D. €780.

Compute the average for $20 \times 5 = 115$

$$\text{Opening inventory} - \frac{115}{100} \times \text{€600} = \text{€690}$$

Multiple Choice Questions (Continued)

2.26 What is the current cost of goods sold?

A. €3,073.

B. €3,241.

C. €3,331.

D. €3,940.

$\text{Current closing inventory} = \frac{115}{130} \times €790 = €699$	
Opening inventory (<i>see</i> 25)	€690
Purchases	3 250
	3 940
Closing inventory	699
Current cost of goods sold	€3 241

Multiple Choice Questions (Continued)

2.27 What is the cost of sales adjustment?

A. €181.

B. €271.

C. €329.

D. €362.

Current cost of sales	€3 241
Historic cost of sales	3 060
Cost of sales adjustment	<u>181</u>

Case Study 2.1

Orange Computer Printers Limited operates two selling and distribution divisions – one in Scotland and one in Canada. With similar market sizes to address, their annual operating performances are identical. However, the Scottish division values its inventory on a FIFO basis, whereas the Canadian division applies a LIFO approach.

The following information is available for each division:

	Units	Unit price €
Sales	25 000	70
Purchases		
April	10 000	35
June	15 000	45
September	4 000	50
Opening inventory	5 000	25
Selling and distribution costs = €575 000		

Required

Prepare the profit and loss accounts for each division and explain the reason for any difference in performance.

Case Study 2.1

Profit and loss account

		Scottish division	
		€000	€000
Sales (25 000 × €70)			1750
<i>Less:</i> Cost of sales			
Opening inventory (5000 × €25)		125	
<i>Add:</i> Purchases			
10 000 × €35	€350 000		
15 000 × €45	675 000		
4 000 × €50	200 000		
29 000	€1 225 000	1 225	
		1 350	
<i>Less:</i> Closing inventory (FIFO)			
4 000 × €50	€200 000		
5 000 × €45	225 000		
9 000	€425 000	425	
			925
Gross profit			825
Selling and distribution costs			575
Net profit			250

Case Study 2.1

Profit and loss account

		Canadian division	
		€000	€000
Sales (25 000 × €70)			1750
Less: Cost of sales			
Opening inventory (5000 × €25)		125	
Add: Purchases			
10 000 × €35	€350 000		
15 000 × €45	675 000		
4 000 × €50	200 000		
29 000	€1 225 000	1 225	
		1 350	
Less: Closing inventory (LIFO)			
5 000 × €25	€125 000		
4 000 × €35	140 000		
9 000	€265 000	265	
			1 085
Gross profit			665
Selling and distribution costs			575
Net profit			90

Case Study 2.3

On 1 October 20x0, Lowland Products Limited commenced trading with the issue of 80,000 Ordinary Shares of 50 cents each. The issue was fully paid up and its proceeds were lodged in the company's bank account at Midwest Bank plc. During the company's first quarter of trading, the following economic actions occurred:

Case Study 2.3 (Continued)

Date

04.10.x0	Paid €4000 for six months' advance rent
10.10.x0	Paid for plant, costing €10 000 and van costing €7000
15.10.x0	Purchased raw materials, costing €8000, on credit
16.10.x0	Paid for van repairs, costing €250
18.10.x0	Paid for raw materials, costing €4000
31.10.x0	Paid wages of €3000 for October, converting half of raw materials Inventory into finished goods
06.11.x0	Purchased office equipment, costing €4500, on credit
08.11.x0	Sold finished goods inventory for €13 000 cash
15.11.x0	Paid €6000 to creditors
21.11.x0	Purchased raw materials costing €7000, on credit
30.11.x0	Paid wages of €3500 for November, converting a further €5000 of raw material inventory into finished goods
01.12.x0	Sold finished goods inventory, costing €5300, for €8600, on credit
05.12.x0	Sold the remainder of finished goods inventory for €7700, on credit
06.12.x0	Paid for raw materials, costing €5400
12.12.x0	Paid €1000 for advertising
15.12.x0	Received credit note for €500 from suppliers following the return of defective raw materials
24.12.x0	Paid wages of €2800 to convert €4200 of raw materials into finished goods
31.12.x0	Received €2500 from debtors

Case Study 2.3 (Continued)

Before final accounts can be prepared for the quarter to 31 December 20x0, additional information has to be taken into consideration.

1. The company has decided to apply the following depreciation policy:
 - a) Plant 15 per cent per annum, reducing-balance method
 - b) Van 25 per cent and office equipment – 25 per cent per annum, straight-line methods
2. One of Lowland's customers has gone into liquidation in early 20x1, owing €1,100, with no prospect of any recovery.
3. At 31 December 20x0, there is an unpaid electricity bill of €480.

Required

1. Prepare the accounting equation as at 30 November 20x0.
2. Prepare the profit and loss account for the quarter to 31 December 20x0.
3. Prepare the balance sheet as at 31 December 20x0.

Case Study 2.3 (Continued)

1 Accounting equation at 30 November 20x0

Plant €10 000 + Van €7 000 + Office equipment €4 500 + Raw material inventory €8 000 + Finished goods inventory €8 500 + Cash at bank €15 250	=	Share capital €40 000 + Profit and loss account €(250) + Creditors €13 500
Total = €53 250		Total = €53 250

Case Study 2.3 (Continued)

2 Profit and Loss Account Three months to 31 December 20x0

	€	€
Sales (13 000 + 8600 + 7700)		29 300
Less: Cost of sales (9000 + 5300 + 3200)	17 500	
Depreciation – plant (note a)	<u>375</u>	
		<u>17 875</u>
Gross profit		11 425
General expenses		
Advertising	1 000	
Bad debts written off	1 100	
Depreciation – office equipment (note b)	281	
Depreciation – van (note c)	438	
Electricity	480	
Rent (note d)	2 000	
Repairs to van	<u>250</u>	
		<u>5 549</u>
Net profit		<u>5 876</u>

Case Study 2.3 (Continued)

3 Balance Sheet as at 31 December 20x0

	€	€
Fixed assets		
Plant (note e)	9 625	
Van (note e)	6 562	
Office equipment (note e)	4 219	
		20 406
Current assets		
Inventories (8 700 + 7 000)	15 700	
Debtors (13 800 – 1 100)	12 700	
Prepayments (note d)	2 000	
Cash at bank	8 550	
		38 950
Less: Current liabilities		
Creditors	13 000	
Accruals (note f)	480	
		13 480
Net current assets		25 470
Net assets		
Represented by:		
Ordinary share capital		40 000
Profit and loss account		5 876
		45 876

Case Study 2.3 (Continued)

Notes

- (a) Depreciation: plant = $(15\% \times \text{€}10\,000) / 4 = \text{€}375$
- (b) Depreciation: office equipment = $(25\% \times \text{€}4\,500) / 4 = \text{€}281$
- (c) Depreciation: van = $(25\% \times \text{€}7\,000) / 4 = \text{€}438$
- (d) Rent (six months) = $\text{€}4\,000 - \text{Prepaid €}2\,000 (1/2) = \text{€}2\,000$
- (e) Book value

	Cost	Depreciation	NBV
	€	€	€
Plant	10 000	375	9 625
Van	7 000	438	6 562
Office equipment	4 500	281	4 219

- (f) Accrued electricity: €480.